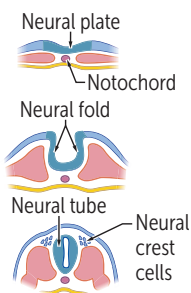


► NEUROLOGY—EMBRYOLOGY

Neural development

Notochord (precursor to nucleus pulposus of intervertebral discs) induces ectoderm to form neuroectoderm → neural plate.

Neural plate gives rise to neural tube and neural crest cells.

Lateral walls of neural tube are divided into alar and basal plates.

Alar plate (dorsal): sensory; induced by bone morphogenetic proteins (BMPs)

Basal plate (ventral): motor; induced by sonic hedgehog (SHH)

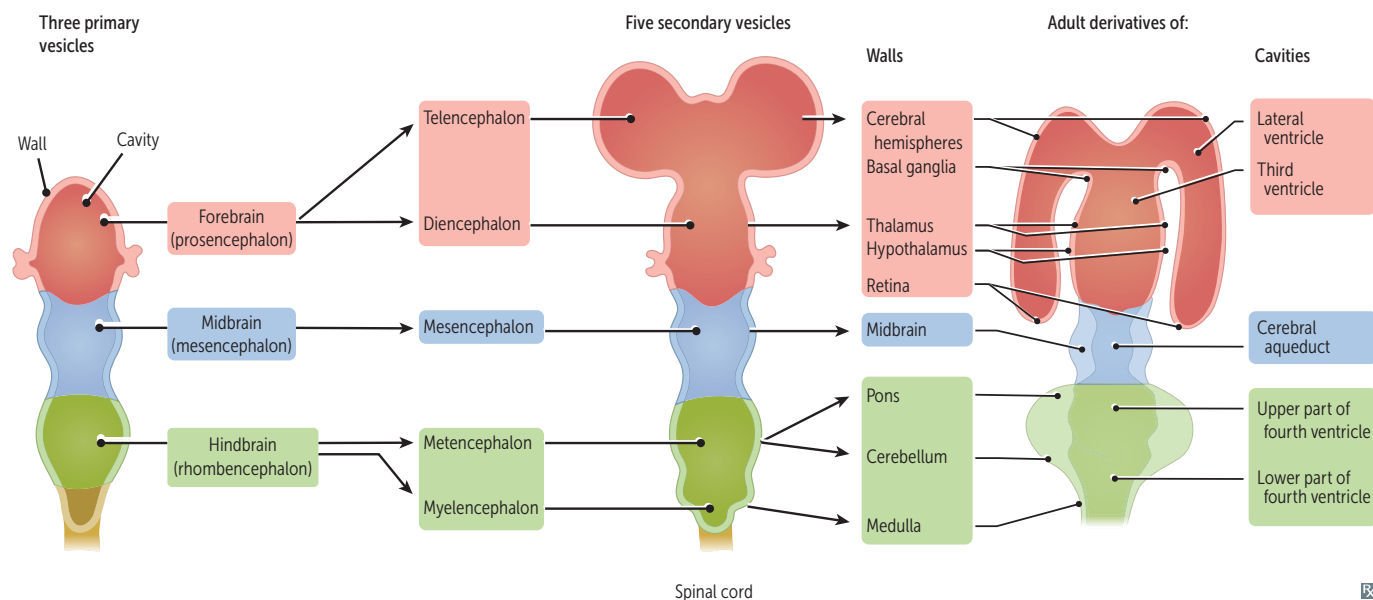
Same orientation as spinal cord

Homeobox (*HOX*) genes regulate neural tube segmentation, cranial-caudal differentiation.

Mutations → syndactyly (limbs), hypospadias (urogenital).

Regionalization of neural tube

Telencephalon is the 1st part. Diencephalon is the 2nd part. The rest are arranged alphabetically: mesencephalon, metencephalon, myelencephalon.

**Central and peripheral nervous systems origins**

Neuroepithelia in neural tube—CNS neurons, CNS glial cells (astrocytes, oligodendrocytes, ependymal cells).

Neural crest—PNS neurons (dorsal root ganglia, autonomic ganglia [sympathetic, parasympathetic, enteric]), PNS glial cells (Schwann cells, satellite cells), adrenal medulla.

Mesoderm—microglia (specialized macrophages).

Neural tube defects

Failure of the neural tube to close completely by week 4 of development, associated with maternal folate deficiency or exposure to teratogens such as valproate and carbamazepine during pregnancy. Diagnosis: ultrasound, maternal serum AFP and/or amniotic fluid AChE (↑ in open NTDs).

Spinal dysraphism**Spina bifida occulta**

Closed NTD. Failure of caudal neural tube to close, but no herniation. Dura is intact. Usually seen at lower vertebral levels. Associated with tuft of hair or skin dimple at level of bony defect.

Meningocele

Open NTD. Meninges (but no neural tissue) herniate through bony defect.

Myelomeningocele

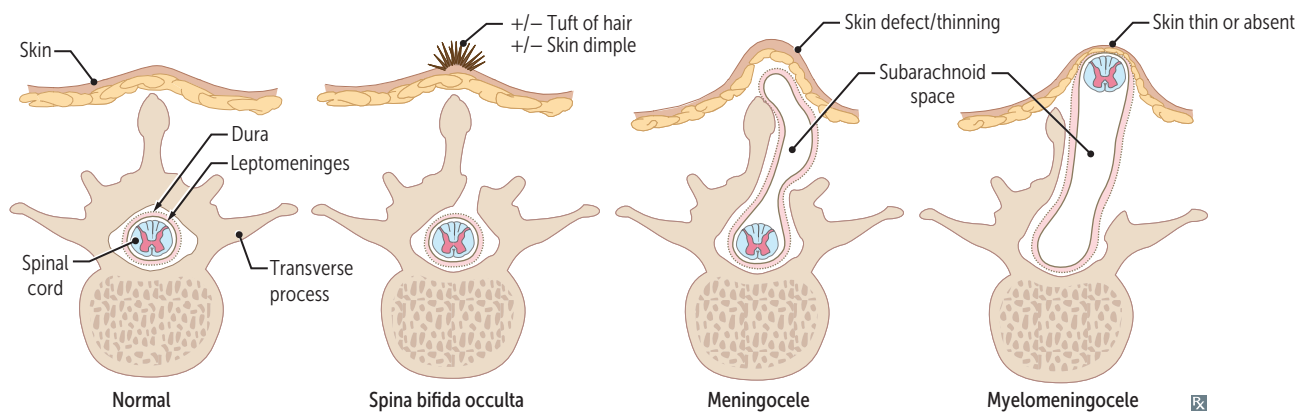
Open NTD. Meninges and neural tissue (eg, cauda equina) herniate through bony defect.

Myeloschisis

Open NTD. Exposed, unfused neural tissue without skin/meningeal covering.

Cranial dysraphism**Anencephaly**

Open NTD. Failure of rostral neuropore to close → no forebrain, open calvarium. Often presents with polyhydramnios (↓ fetal swallowing due to lack of neural control).

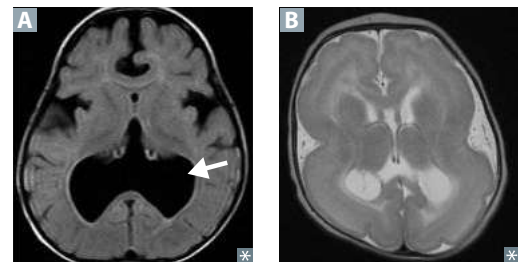
**Brain malformations**

Often incompatible with postnatal life. Survivors may be profoundly disabled.

Holoprosencephaly

Failure of forebrain (prosencephalon) to divide into 2 cerebral hemispheres; developmental field defect usually occurring at weeks 3–4 of development. Associated with *SHH* mutations. May be seen in Patau syndrome (trisomy 13), fetal alcohol syndrome.

Presents with midline defects: monoventricle **A**, fused basal ganglia, cleft lip/palate, hypotelorism, cyclopia, proboscis. ↑ risk for pituitary dysfunction (eg, diabetes insipidus).

**Lissencephaly**

Failure of neuronal migration → smooth brain surface that lacks sulci and gyri **B**.

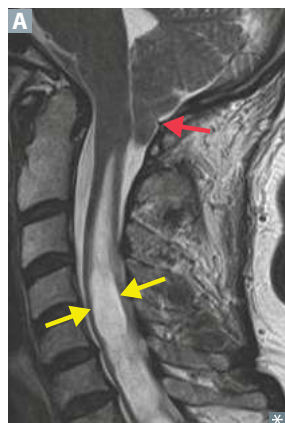
Presents with dysphagia, seizures, microcephaly, facial anomalies.

Posterior fossa malformations

- Chiari I malformation** Downward displacement of cerebellar **tonsils** through foramen magnum (**1** structure) **A**. Usually asymptomatic in childhood, manifests in adulthood with headaches and cerebellar symptoms. Associated with spinal cord cavitations (eg, syringomyelia).
- Chiari II malformation** Downward displacement of **cerebellum** (vermis and tonsils) and **medulla** (**2** structures) through foramen magnum → noncommunicating hydrocephalus. More severe than Chiari I, usually presents early in life with dysphagia, stridor, apnea, limb weakness. Associated with myelomeningocele (usually lumbosacral).
- Dandy-Walker malformation** Agenesis of cerebellar vermis → cystic enlargement of 4th ventricle (arrow in **B**) that fills the enlarged posterior fossa. Associated with noncommunicating hydrocephalus, spina bifida.



Syringomyelia



Fluid-filled, gliosis-lined cavity within spinal cord (yellow arrows in **A**). Fibers crossing in anterior white commissure (spinothalamic tract) are typically damaged first → “cape-like” loss of pain and temperature sensation in bilateral upper extremities. As lesion expands it may damage anterior horns → lower motor neuron (LMN) findings.

Syrinx (Greek) = tube, as in “**syringe**.”

Most lesions occur between C2 and T9. Usually associated with Chiari I malformation (red arrow in **A**). Less commonly associated with other malformations, infections, tumors, trauma.

